Group 7 Project - Milestone 3 (Due Monday, July 14)

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1. SQL Queries with Descriptions

Home page - Roy & Dan

Queries

1. **SIMPLE QUERY**

Homepage has an autocomplete search bar at the top. Based on user input text, we will use this SQL query to return partial matches to the text, where a match can be either zip code, city, or state. *Note that partially matching on city means partially matching on a CONCAT(City, ', ', State).*

```
(SELECT z.ZipCode, 'ZipCode' AS MatchType
FROM ZipCode z
WHERE z.ZipCode LIKE '<searchTerm>%')
UNION
(SELECT DISTINCT
  CONCAT(City, ', ', State) As CityAndState,
  'City' AS MatchType
FROM ZipCode
WHERE CONCAT(City, ', ', State) ILIKE '%<searchTerm>%')
UNION
(SELECT DISTINCT State, 'State' AS MatchType
FROM ZipCode
WHERE State ILIKE '%<searchTerm>%');
```

2. **COMPLEX QUERY**

On the homepage, we plan to feature ~3 cities that would be interesting to the user. This query returns the top 3 cities with population >= 100,000 that have both the lowest average home prices and the lowest measures of our target health risks.

```
WITH CityPriceRank AS (
```

```
  SELECT
    City,
    State,
    SUM(Population) AS Population,
    AVG(lm.MedianListingPrice) AS AvgListingPrice,
    RANK() OVER
      (ORDER BY AVG(lm.MedianListingPrice) ASC) AS PriceRank
  FROM ZipCode z JOIN LocalMarket lm ON z.ZipCode = lm.ZipCode
  GROUP BY City, State
),
CityHealthRank AS (
  SELECT
    City,
    State,
    SUM(Population) AS Population,
    AVG(hm.Ratio) AS AvgHealthMeasure,
    RANK() OVER
      (ORDER BY AVG(hm.Ratio) ASC) AS HealthRank
  FROM ZipCode z JOIN HealthMeasure hm ON z.ZipCode = hm.ZipCode
  GROUP BY City, State
)
SELECT
  cpr.City, cpr.State, cpr.Population,
  AvgListingPrice, AvgHealthMeasure
FROM CityPriceRank cpr
  JOIN CityHealthRank chr
  ON cpr.City = chr.City AND cpr.State = chr.State
WHERE cpr.Population >= 100000
ORDER BY (PriceRank + HealthRank) ASC
LIMIT 3;
```

3. **COMPLEX QUERY**

On the homepage, we plan to feature ~3 cities / zipCode that would be interesting to the user. This query returns the top 3 Zip Codes with population >= 2000 that have both good social service resources and healthcare resources.

```
WITH facility_counts AS (
    -- Count public safety and health facilities per zipcode
    SELECT
        zipcode,
        COUNT(DISTINCT hospital_id) as hospital_count,
        COUNT(DISTINCT police_id) as police_count,
        COUNT(DISTINCT firefighter_id) as firefighter_count,
        (COUNT(DISTINCT hospital_id) + COUNT(DISTINCT police_id) +
COUNT(DISTINCT firefighter_id)) as total_facilities
    FROM (
            SELECT zipcode, CAST(id AS TEXT) as hospital_id, NULL
as police_id, NULL as firefighter_id
            FROM hospitals
            UNION ALL
            SELECT zipcode, NULL, CAST(id AS TEXT) as police_id,
```

```
NULL
            FROM policestations
            UNION ALL
            SELECT zipcode, NULL, NULL, CAST(dptid AS TEXT) as
firefighter_id
            FROM firefighterstations
        ) all_facilities
    GROUP BY zipcode
),
     health_metrics AS (
         -- Calculate health outcome scores (higher = worse health
outcomes)
         SELECT
             zipcode,
             AVG(CASE
                     WHEN measure LIKE '%Obesity%' THEN ratio * 100
                     WHEN measure LIKE '%asthma%' THEN ratio * 100
                     WHEN measure LIKE '%Depression%' THEN ratio *
100
                     WHEN measure LIKE '%Housing insecurity%' THEN
ratio * 100
                     WHEN measure LIKE '%socially isolated%' THEN
ratio * 100
                     WHEN measure LIKE '%routine checkup%' THEN (1
- ratio) * 100  -- Inverse: lower checkup rate = worse
                     ELSE NULL
                 END) as avg_poor_health_ratio,
             COUNT(DISTINCT measure) as health_measures_count,
             MAX(totalpopulation) as population
         FROM healthmeasure
         WHERE measure IN (
                           'Obesity among adults',
                           'Current asthma among adults',
                           'Depression among adults',
                           'Housing insecurity in the past 12
months among adults',
                           'Feeling socially isolated among
adults',
                           'Visits to doctor for routine checkup
within the past year among adults'
             )
         GROUP BY zipcode
     ),
     income_data AS (
         -- Get income data for additional context
         SELECT
             zipcode,
             meanincome
         FROM householdincome
     ),
     combined_metrics AS (
```

```
        -- Combine facility counts and health metrics
        SELECT
            fc.zipcode,
            fc.hospital_count,
            fc.police_count,
            fc.firefighter_count,
            fc.total_facilities,
            hm.avg_poor_health_ratio,
            hm.health_measures_count,
            hm.population,
            id.meanincome,
            -- Normalize facility count (0-1 scale, 0 =
underserved, 1 = well served)
            CASE
                WHEN fc.total_facilities = 0 THEN 0.0
                WHEN fc.total_facilities >= 10 THEN 1.0
                ELSE fc.total_facilities / 10.0
                END as staffing_service_score,
            -- Normalize health outcomes (0-1 scale, 0 = poor
health, 1 = good health)
            CASE
                WHEN hm.avg_poor_health_ratio IS NULL THEN 0.0
                WHEN hm.avg_poor_health_ratio <= 10 THEN 1.0
                WHEN hm.avg_poor_health_ratio >= 50 THEN 0.0
                ELSE 1.0 - ((hm.avg_poor_health_ratio - 10) /
40.0)
                END as health_service_score
        FROM facility_counts fc
                 LEFT JOIN health_metrics hm ON fc.zipcode =
hm.zipcode
                 LEFT JOIN income_data id ON fc.zipcode =
id.zipcode
        WHERE hm.population >= 2000
          AND hm.health_measures_count >= 3  -- At least 3 health
measures available
    ),
    underserved_ranking AS (
        -- Calculate service level score and rank (higher score =
better served)
        SELECT
            *,
            (staffing_service_score * 0.1 + health_service_score *
0.1) as service_level_score,
            ROW_NUMBER() OVER (ORDER BY (staffing_service_score *
0.1 + health_service_score * 0.1) ASC) as rank
        FROM combined_metrics
        WHERE staffing_service_score < 1.0 OR health_service_score
< 1.0
    )
SELECT
    zipcode,
```

```
    hospital_count,
    police_count,
    firefighter_count,
    total_facilities,
    ROUND(avg_poor_health_ratio::numeric, 2) as
avg_poor_health_ratio,
    health_measures_count,
    population,
    meanincome,
    ROUND(staffing_service_score::numeric, 3) as
staffing_service_score,
    ROUND(health_service_score::numeric, 3) as
health_service_score,
    ROUND(service_level_score::numeric, 3) as service_level_score,
    rank
FROM underserved_ranking
ORDER BY service_level_score DESC
LIMIT 4;
```

Search results page - Dan & Filmon

Queries

4. **COMPLEX QUERY**

This query returns a list of ZIP codes with summary data about each of them—filtering that list based on multiple filter inputs available to the user in the UI. Each row will be used to display a (1) result card using that summary data and (2) dot on the interactive map using latitude and longitude.

Available user filters:

- State
- City in state (optional)
- Housing price range (min and max)
- Mean income range (min and max)
- Health measure choice (1 of 6 options)
- Health measure threshold (max)
- Police departments threshold (min)
  Police officer count threshold (min)
- Hospitals threshold (min)
  Firefighter stations threshold (min)
- Firefighter count threshold (min)
- Childcare centers threshold (min)
- Total population range (min and max)

NOTE: This query is very slow right now when search results are for an entire state and all filters are minimally restrictive. For example, a query for “NY” where all filters are set to 0s for minimums and really high numbers for maximums takes approximately **2 minutes** to execute.

```
WITH ZipCodesInCityOrState AS (
    SELECT *
    FROM ZipCode
    WHERE State = :state
      AND (:city IS NULL OR City = :city)
),

LatestMarket AS (
    SELECT lm.ZipCode, lm.MedianListingPrice
    FROM LocalMarket lm
    JOIN (
        SELECT lm2.ZipCode, MAX(lm2.MonthDate) AS LatestMonth
        FROM LocalMarket lm2
        JOIN ZipCodesInCityOrState z ON lm2.ZipCode = z.ZipCode
        GROUP BY lm2.ZipCode
    ) latest ON lm.ZipCode = latest.ZipCode AND lm.MonthDate =
latest.LatestMonth
),

IncomeStats AS (
    SELECT hi.ZipCode, hi.MeanIncome
    FROM HouseholdIncome hi
    JOIN ZipCodesInCityOrState z ON hi.ZipCode = z.ZipCode
),

HealthStats AS (
    SELECT hm.ZipCode, hm.Ratio AS HealthRatio
    FROM HealthMeasure hm
    JOIN ZipCodesInCityOrState z ON hm.ZipCode = z.ZipCode
    WHERE hm.Measure = :health_measure
),

PoliceStats AS (
    SELECT ps.ZipCode,
           COUNT(*) AS PoliceDeptCount,
           SUM(TotalActiveOfficers) AS PoliceOfficerCount
    FROM PoliceStations ps
    JOIN ZipCodesInCityOrState z ON ps.ZipCode = z.ZipCode
    GROUP BY ps.ZipCode
),

HospitalStats AS (
    SELECT h.ZipCode, COUNT(*) AS HospitalCount
    FROM Hospitals h
    JOIN ZipCodesInCityOrState z ON h.ZipCode = z.ZipCode
    GROUP BY h.ZipCode
),

FireStats AS (
    SELECT f.ZipCode,
           SUM(NumberOfStations) AS FireStationCount,
```

```sql
            SUM(ActivePersonnel) AS FirefighterCount
    FROM FirefighterStations f
    JOIN ZipCodesInCityOrState z ON f.ZipCode = z.ZipCode
    GROUP BY f.ZipCode
),

ChildcareStats AS (
    SELECT c.ZipCode, COUNT(*) AS ChildcareCount
    FROM ChildcareCenters c
    JOIN ZipCodesInCityOrState z ON c.ZipCode = z.ZipCode
    GROUP BY c.ZipCode
)

SELECT
    z.ZipCode,
    z.City,
    z.State,
    COALESCE(z.Population, 0) AS Population,
    lm.MedianListingPrice     AS MedianPrice,
    ins.MeanIncome            AS MeanIncome,
    hes.HealthRatio           AS HealthRatio,
    ps.PoliceDeptCount        AS PoliceDepartmentsCount,
    ps.PoliceOfficerCount     AS NumPoliceOfficersCount,
    hos.HospitalCount         AS HospitalsCount,
    fs.FireStationCount       AS FireStationsCount,
    fs.FirefighterCount       AS FirefightersCount,
    cs.ChildcareCount         AS ChildcareCentersCount

FROM ZipCodesInCityOrState z
LEFT OUTER JOIN LatestMarket      lm  ON z.ZipCode = lm.ZipCode
LEFT OUTER JOIN IncomeStats       ins ON z.ZipCode = ins.ZipCode
LEFT OUTER JOIN HealthStats       hes ON z.ZipCode = hes.ZipCode
LEFT OUTER JOIN PoliceStats       ps  ON z.ZipCode = ps.ZipCode
LEFT OUTER JOIN HospitalStats     hos ON z.ZipCode = hos.ZipCode
LEFT OUTER JOIN FireStats         fs  ON z.ZipCode = fs.ZipCode
LEFT OUTER JOIN ChildcareStats    cs  ON z.ZipCode = cs.ZipCode

WHERE COALESCE(lm.MedianListingPrice, 0) >= :min_price
  AND COALESCE(lm.MedianListingPrice, 0) <= :max_price
  AND COALESCE(ins.MeanIncome, 0) >= :min_income
  AND COALESCE(ins.MeanIncome, 0) <= :max_income
  AND COALESCE(hes.HealthRatio, 1) <= :max_health_ratio
  AND COALESCE(ps.PoliceDeptCount, 0) >= :min_police_depts
  AND COALESCE(ps.PoliceOfficerCount, 0) >= :min_police_officers
  AND COALESCE(hos.HospitalCount, 0) >= :min_hospitals
  AND COALESCE(fs.FireStationCount, 0) >= :min_fire_stations
  AND COALESCE(fs.FirefighterCount, 0) >= :min_firefighters
  AND COALESCE(cs.ChildcareCount, 0) >= :min_childcare_centers
  AND COALESCE(z.Population, 0) >= :min_population
  AND COALESCE(z.Population, 0) <= :max_population;
```

5. **SIMPLE QUERY**

This query simply returns the list of all available health measures, which we will use for a dropdown filter option in the search results page.

```
SELECT DISTINCT Measure FROM HealthMeasure ORDER BY Measure;
```

Location detail page - Filmon & John

Queries

6. **SIMPLE QUERY**

This query returns the full description required to fill the data in the location page for a given zipcode ZIP. (Note: we can reuse it for the Comparison Page)

```
WITH firetable AS (
  SELECT
    ZipCode,
    SUM(ActivePersonnel) AS Firefighters,
    COUNT(DISTINCT DptId) AS FirefighterDepartments
  FROM FirefighterStations
  GROUP BY ZipCode
),
policetable AS (
  SELECT
    ZipCode,
    SUM(TotalActiveOfficers) AS PoliceOfficers,
    COUNT(DISTINCT Id) AS PoliceStations
  FROM PoliceStations
  GROUP BY ZipCode
)

SELECT
  z.ZipCode,
  z.City,
  z.State,
  z.Population,
  hi.MeanIncome,
  lm.MedianListingPrice,
  lm.ActiveListingCount,
  pt.PoliceStations,
  pt.PoliceOfficers,
  COUNT(DISTINCT h.Id) AS Hospitals,
  COUNT(DISTINCT cc.Id) AS ChildcareCenters,
  ft.FirefighterDepartments,
  ft.Firefighters
FROM ZipCode z
LEFT JOIN HouseholdIncome hi ON z.ZipCode = hi.ZipCode
```

```
LEFT JOIN LocalMarket lm ON z.ZipCode = lm.ZipCode
LEFT JOIN policetable pt ON z.ZipCode = pt.ZipCode
LEFT JOIN firetable ft ON z.ZipCode = ft.ZipCode
LEFT JOIN Hospitals h ON z.ZipCode = h.ZipCode
LEFT JOIN ChildcareCenters cc ON z.ZipCode = cc.ZipCode
WHERE z.ZipCode = ZIP
GROUP BY z.ZipCode, z.City, z.State, z.Population,
         hi.MeanIncome, lm.MedianListingPrice,
lm.ActiveListingCount,
         pt.PoliceStations, pt.PoliceOfficers,
         ft.FirefighterDepartments, ft.Firefighters,
         lm.monthdate
ORDER BY lm.monthdate DESC
LIMIT 1;
```

7. **SIMPLE QUERY**

This query returns market trends Over Time (Median Listing Price and month date) for a ZIP Code. It can be mapped into a chart component (Note: we can reuse it for the Comparison Page)

```
SELECT MonthDate, MedianListingPrice
FROM LocalMarket
WHERE ZipCode = ZIP
ORDER BY MonthDate;
```

8. **SIMPLE QUERY**

This query returns community health properties (asthma, obesity, social isolation etc.) along with the total number of hospitals and hospital beds per population for a given zipcode ZIP. (Note: we can reuse it for the Comparison Page)

```
WITH hospitalTable AS (
  SELECT
    z.ZipCode,
    COUNT(DISTINCT h.Id) AS HospitalCount,
    CASE
      WHEN COUNT(DISTINCT h.Id) > 0
           AND z.Population > 0
           THEN ROUND(z.Population / COUNT(DISTINCT h.Id), 2)
      ELSE 0
      END AS HospitalToPopulationRatio,
    CASE
      WHEN SUM(h.numofbeds) > 0
           AND z.Population > 0
           THEN ROUND(z.Population / SUM(h.numofbeds), 2)
      ELSE 0
      END AS BedsToPopulationRatio
  FROM ZipCode z
  LEFT JOIN Hospitals h ON z.ZipCode = h.ZipCode
  GROUP BY z.ZipCode, z.Population
)
```

```
SELECT
  MAX(CASE WHEN hm.Measure = 'Current asthma among adults'
      THEN hm.Ratio END) AS AsthmaRate,
  MAX(CASE WHEN hm.Measure = 'Obesity among adults'
      THEN hm.Ratio END) AS ObesityRate,
  MAX(CASE WHEN hm.Measure = 'Visits to doctor for routine checkup
within the past year among adults'
      THEN hm.Ratio END) AS DoctorVisitsRate,
  MAX(CASE WHEN hm.Measure = 'Depression among adults'
      THEN hm.Ratio END) AS DepressionRate,
  MAX(CASE WHEN hm.Measure = 'Housing insecurity in the past 12
months among adults'
      THEN hm.Ratio END) AS HousingInsecurityRate,
  MAX(CASE WHEN hm.Measure = 'Feeling socially isolated among
adults'
      THEN hm.Ratio END) AS SocialIsolationRate,
  ht.HospitalCount,
  ht.HospitalToPopulationRatio,
  ht.BedsToPopulationRatio
FROM ZipCode z
LEFT JOIN HealthMeasure hm ON z.ZipCode = hm.ZipCode
LEFT JOIN hospitalTable ht ON z.ZipCode = ht.ZipCode
WHERE z.ZipCode = '77002'
GROUP BY z.ZipCode,
         z.Population,
         ht.HospitalCount,
         ht.HospitalToPopulationRatio,
         ht.BedsToPopulationRatio;
```

9. **Complex query - roughly 8 seconds**

This query finds the 4 most similar **zipcodes** for the given ZIP code. It is measured as the sum of absolute difference between different attributes such as population, meanincome, average listing price, services (police, firefighter, childcare), and health measures. Each attribute is assigned a weighted value between 0 and 1 in calculations.

```
WITH firetable AS (
  SELECT
    ZipCode,
    SUM(ActivePersonnel) AS Firefighters,
    COUNT(DISTINCT DptId) AS FirefighterDepartments
  FROM FirefighterStations
  GROUP BY ZipCode
),

policetable AS (
  SELECT
    ZipCode,
    SUM(TotalActiveOfficers) AS PoliceOfficers,
    COUNT(DISTINCT Id) AS PoliceStations
  FROM PoliceStations
```

```
  GROUP BY ZipCode
),

listingtable AS (
    SELECT
        zipcode,
        AVG(medianlistingprice) as avgmedianlistingprice
    FROM localmarket
    GROUP BY zipcode
),

current_zipcode AS (
  SELECT
    z.ZipCode,
    z.Population,
    hi.MeanIncome,
    lt.avgmedianlistingprice,
    pt.PoliceStations,
    pt.PoliceOfficers,
    COUNT(DISTINCT h.Id) AS Hospitals,
    COUNT(DISTINCT cc.Id) AS ChildcareCenters,
    ft.FirefighterDepartments,
    ft.Firefighters,
    MAX(CASE WHEN hm.Measure = 'Current asthma among adults'
        THEN hm.Ratio END) AS AsthmaRate,
    MAX(CASE WHEN hm.Measure = 'Obesity among adults'
        THEN hm.Ratio END) AS ObesityRate,
    MAX(CASE WHEN hm.Measure = 'Visits to doctor for routine
checkup within the past year among adults'
        THEN hm.Ratio END) AS DoctorVisitsRate,
    MAX(CASE WHEN hm.Measure = 'Depression among adults'
        THEN hm.Ratio END) AS DepressionRate,
    MAX(CASE WHEN hm.Measure = 'Housing insecurity in the past 12
months among adults'
        THEN hm.Ratio END) AS HousingInsecurityRate,
    MAX(CASE WHEN hm.Measure = 'Feeling socially isolated among
adults'
        THEN hm.Ratio END) AS SocialIsolationRate
  FROM ZipCode z
  LEFT JOIN HouseholdIncome hi ON z.ZipCode = hi.ZipCode
  LEFT JOIN HealthMeasure hm ON z.ZipCode = hm.ZipCode
  LEFT JOIN policetable pt ON z.ZipCode = pt.ZipCode
  LEFT JOIN firetable ft ON z.ZipCode = ft.ZipCode
  LEFT JOIN Hospitals h ON z.ZipCode = h.ZipCode
  LEFT JOIN ChildcareCenters cc ON z.ZipCode = cc.ZipCode
  LEFT JOIN listingtable lt ON z.zipcode = lt.zipcode
  WHERE z.ZipCode = '48416'
  GROUP BY z.ZipCode, z.Population, hi.MeanIncome,
pt.PoliceStations, pt.PoliceOfficers, ft.FirefighterDepartments,
ft.Firefighters, lt.avgmedianlistingprice
),
```

```
all_zipcodes AS (
  SELECT
    z.ZipCode,
    z.Population,
    hi.MeanIncome,
    lt.avgmedianlistingprice,
    pt.PoliceStations,
    pt.PoliceOfficers,
    COUNT(DISTINCT h.Id) AS Hospitals,
    COUNT(DISTINCT cc.Id) AS ChildcareCenters,
    ft.FirefighterDepartments,
    ft.Firefighters,
    MAX(CASE WHEN hm.Measure = 'Current asthma among adults'
        THEN hm.Ratio END) AS AsthmaRate,
    MAX(CASE WHEN hm.Measure = 'Obesity among adults'
        THEN hm.Ratio END) AS ObesityRate,
    MAX(CASE WHEN hm.Measure = 'Visits to doctor for routine
checkup within the past year among adults'
        THEN hm.Ratio END) AS DoctorVisitsRate,
    MAX(CASE WHEN hm.Measure = 'Depression among adults'
        THEN hm.Ratio END) AS DepressionRate,
    MAX(CASE WHEN hm.Measure = 'Housing insecurity in the past 12
months among adults'
        THEN hm.Ratio END) AS HousingInsecurityRate,
    MAX(CASE WHEN hm.Measure = 'Feeling socially isolated among
adults'
        THEN hm.Ratio END) AS SocialIsolationRate
  FROM ZipCode z
  LEFT JOIN HouseholdIncome hi ON z.ZipCode = hi.ZipCode
  LEFT JOIN HealthMeasure hm ON z.ZipCode = hm.ZipCode
  LEFT JOIN policetable pt ON z.ZipCode = pt.ZipCode
  LEFT JOIN firetable ft ON z.ZipCode = ft.ZipCode
  LEFT JOIN Hospitals h ON z.ZipCode = h.ZipCode
  LEFT JOIN ChildcareCenters cc ON z.ZipCode = cc.ZipCode
  LEFT JOIN listingtable lt ON z.zipcode = lt.zipcode
  WHERE z.ZipCode <> '48416'
  GROUP BY z.ZipCode, z.Population, hi.MeanIncome,
pt.PoliceStations, pt.PoliceOfficers, ft.FirefighterDepartments,
ft.Firefighters, lt.avgmedianlistingprice
)

SELECT
  az.ZipCode,
  ROUND(
    ABS(COALESCE(az.Population, 0) - COALESCE(cz.Population, 0)) *
0.05 +
    ABS(COALESCE(az.MeanIncome, 0) - COALESCE(cz.MeanIncome, 0)) *
0.1 +
    ABS(COALESCE(az.avgmedianlistingprice, 0) -
COALESCE(cz.avgmedianlistingprice, 0)) * 0.1 +
```

```
    ABS(COALESCE(az.PoliceStations, 0) -
COALESCE(cz.PoliceStations, 0)) * 0.05 +
    ABS(COALESCE(az.PoliceOfficers, 0) -
COALESCE(cz.PoliceOfficers, 0)) * 0.05 +
    ABS(COALESCE(az.Hospitals, 0) - COALESCE(cz.Hospitals, 0)) *
0.1 +
    ABS(COALESCE(az.ChildcareCenters, 0) -
COALESCE(cz.ChildcareCenters, 0)) * 0.1 +
    ABS(COALESCE(az.FirefighterDepartments, 0) -
COALESCE(cz.FirefighterDepartments, 0)) * 0.05 +
    ABS(COALESCE(az.Firefighters, 0) - COALESCE(cz.Firefighters,
0)) * 0.05 +
    ABS(COALESCE(az.AsthmaRate, 0) - COALESCE(cz.AsthmaRate, 0)) *
0.05 +
    ABS(COALESCE(az.ObesityRate, 0) - COALESCE(cz.ObesityRate, 0))
* 0.05 +
    ABS(COALESCE(az.DoctorVisitsRate, 0) -
COALESCE(cz.DoctorVisitsRate, 0)) * 0.05 +
    ABS(COALESCE(az.DepressionRate, 0) -
COALESCE(cz.DepressionRate, 0)) * 0.05 +
    ABS(COALESCE(az.HousingInsecurityRate, 0) -
COALESCE(cz.HousingInsecurityRate, 0)) * 0.05 +
    ABS(COALESCE(az.SocialIsolationRate, 0) -
COALESCE(cz.SocialIsolationRate, 0)) * 0.05
  ) AS similarity_score
FROM all_zipcodes az
JOIN current_zipcode cz ON TRUE
ORDER BY similarity_score ASC
LIMIT 4;
```

Comparison page - John & Roy

Queries

10. **Complex Query**

This query returns the aggregation of the number of hospitals,police stations and firefighter stations, Income, Population, and health Data for the query comparison result.

```
With FF_list AS (
    SELECT
        z.zipcode,
        z.City,
        z.State,
        z.Population,
        hi.MeanIncome,
        COUNT(DISTINCT fs.DptId) AS FirefighterDepartments,
        SUM(fs.ActivePersonnel) AS Firefighters
    FROM ZipCode z
```

```
            LEFT JOIN HouseholdIncome hi ON z.ZipCode = hi.ZipCode
            LEFT JOIN FirefighterStations fs ON z.ZipCode =
fs.ZipCode
    WHERE z.City ilike 'Houston' and z.State ilike 'TX'
    GROUP BY z.ZipCode, z.City, z.State, z.Population,
hi.MeanIncome
),
POLICE_list AS (
    SELECT
        z.zipcode,
        z.City,
        z.State,
        COUNT(DISTINCT ps.Id) AS PoliceStations,
        SUM(ps.TotalActiveOfficers) AS PoliceOfficers
    FROM ZipCode z
            LEFT JOIN PoliceStations ps ON z.ZipCode = ps.ZipCode
    WHERE z.City ilike 'Houston' and z.State ilike 'TX'
    GROUP BY z.ZipCode, z.City, z.State
),
CC_list AS (
    SELECT
        z.zipcode,
        z.City,
        z.State,
        COUNT(DISTINCT cc.Id) AS NumChildCareCenter
    FROM ZipCode z
            LEFT JOIN ChildcareCenters cc ON z.ZipCode =
cc.ZipCode
    WHERE z.City ilike 'Houston' and z.State ilike 'TX'
    GROUP BY z.ZipCode, z.City, z.State
),
H_list AS (
    SELECT
        z.zipcode,
        z.City,
        z.State,
        COUNT(DISTINCT h.Id) AS Hospitals
    FROM ZipCode z
            LEFT JOIN hospitals h ON z.ZipCode = h.ZipCode
    WHERE z.City ilike 'Houston' and z.State ilike 'TX'
    GROUP BY z.ZipCode, z.City, z.State
),
HM_list AS (
    SELECT
        z.zipcode,
        z.City,
        z.State,
        hm.ratio AS ObesityRate
    FROM ZipCode z
            LEFT JOIN healthmeasure hm ON z.ZipCode = hm.ZipCode
    WHERE z.City ilike 'Houston' and z.State ilike 'TX' and
```

```
hm.measure like 'Obesity among adults'
)

SELECT FF.City,
       FF.State,
       sum(ff.population) AS TotalPopulation,
       AVG(ff.meanincome) AS MeanIncome,
       sum(FF.FirefighterDepartments) AS NumPoliceDept,
       sum(FF.Firefighters) AS NumPoliceOfficer,
       sum(CC.NumChildCareCenter) AS NumChildCareCenter,
       AVG(HM.ObesityRate) AS ObesityRate
From FF_list FF
             LEFT JOIN POLICE_list PP ON FF.zipcode = PP.zipcode
             LEFT JOIN CC_list CC ON FF.zipcode = CC.zipcode
             LEFT JOIN H_list H ON FF.zipcode = H.zipcode
             LEFT JOIN HM_list HM ON FF.zipcode = HM.zipcode
GROUP BY FF.city, FF.state
```

11. **Simple Query**

This query simply returns the growth rate of the desired city. We will be able to run this query twice for the comparison. Currently. The query is setup to return the growth rate of home price over 3 years.

```
WITH CityGrowRateOld AS (
    SELECT
        Z.city,
        Z.state,
        AVG(lm.MedianListingPrice) AS AvgListingPriceOld
    FROM ZipCode z JOIN LocalMarket lm ON z.ZipCode = lm.ZipCode
    WHERE lm.monthdate like '2022%'
          And Z.City ilike 'Houston'
          And Z.state ilike 'TX'
    GROUP BY Z.city, Z.state
),
     CityGrowRateNew AS (
         SELECT
             Z.city,
             Z.state,
             AVG(lm.MedianListingPrice) AS AvgListingPriceNew
         FROM ZipCode z JOIN LocalMarket lm ON z.ZipCode =
lm.ZipCode
         WHERE lm.monthdate like '2025%'
           And Z.city ilike 'Houston'
           And Z.state ilike 'TX'
         GROUP BY Z.City, Z.State
     )

SELECT
    CO.city,
```

```
    CO.state,
    ROUND(((AvgListingPriceNew - AvgListingPriceOld) /
AvgListingPriceOld * 100), 2)::TEXT || '%' AS GrowthRate3Year
FROM CityGrowRateOld CO
         NATURAL JOIN CityGrowRateNew CN;
```

12. Complex query - roughly 15 seconds

This query finds the similar cities relative to the current city. A similarity calculation is used to determine the most similar city. The query will first classify each city into different category and then filter out similar city based on population to narrow down the search. We then calculate the similarity score based on the query data. The current run time takes roughly 15 seconds.

```
With target_city AS (
    SELECT Z.city,
           Z.state,
           sum(Z.population) AS TotalPopulation,
           CASE
               WHEN sum(z.population) >= 6350 THEN 4
               WHEN sum(z.population) >= 635 THEN 3
               WHEN sum(z.population) >= 165 THEN 2
               ELSE 1 END AS PopulationGrade
    FROM zipcode Z
    WHERE Z.city ilike :city and Z.state ilike :state
    GROUP BY Z.city, Z.state
    LIMIT 1
),

city_pop AS (
    SELECT z.city,
           z.state,
           sum(z.population) AS TotalPopulation,
           CASE
               WHEN sum(z.population) >= 6350 THEN 4
               WHEN sum(z.population) >= 635 THEN 3
               WHEN sum(z.population) >= 165 THEN 2
               ELSE 1 END AS PopulationGrade
    FROM zipcode Z
    GROUP BY z.city, z.state
    ORDER BY TotalPopulation DESC
),

similar_pop AS (
    SELECT C.city, C.state, C.TotalPopulation
    FROM city_pop C JOIN target_city TC ON C.PopulationGrade =
TC.PopulationGrade
    --WHERE C.city != :city and C.state != :state
    limit 20
),
```

```
firetable AS (
  SELECT
    Z.city,
    Z.state,
    SUM(F.ActivePersonnel) AS Firefighters,
    COUNT(DISTINCT F.DptId) AS FirefighterDepartments
  FROM zipcode Z JOIN FirefighterStations F on Z.zipcode =
F.zipcode
  GROUP BY Z.city, Z.state
),

policetable AS (
  SELECT
      Z.city,
      Z.state,
      SUM(TotalActiveOfficers) AS PoliceOfficers,
      COUNT(DISTINCT Id) AS PoliceStations
  FROM zipcode Z JOIN PoliceStations P on Z.zipcode = P.zipcode
  GROUP BY Z.city, Z.state
),

listingtable AS (
    SELECT
        Z.city,
        z.state,
        AVG(medianlistingprice) as avgmedianlistingprice
    FROM zipcode Z JOIN localmarket lm on Z.zipcode = lm.zipcode
    GROUP BY z.city, z.state
),

incometable AS (
    SELECT
        Z.city,
        Z.state,
        avg(Hi.meanincome) AS MeanIncome
    FROM zipcode Z JOIN householdincome HI on Z.zipcode =
HI.zipcode
    GROUP BY Z.city, Z.state
),

hospitaltable AS (
    SELECT
        Z.city,
        Z.state,
        COUNT(DISTINCT h.Id) AS NumHospitals
    FROM zipcode Z JOIN hospitals H on Z.zipcode = H.zipcode
    GROUP BY Z.city, Z.state
),

childtable AS (
    SELECT
```

```
        Z.city,
        Z.state,
        COUNT(DISTINCT cc.Id) AS NumChildcareCenters
    FROM zipcode Z JOIN childcarecenters cc on Z.zipcode =
cc.zipcode
    GROUP BY Z.city, Z.state
),

current_city AS (
    SELECT
        TC.city,
        TC.state,
        TC.TotalPopulation,
        hi.MeanIncome,
        lt.avgmedianlistingprice,
        pt.PoliceStations,
        pt.PoliceOfficers,
        h.NumHospitals,
        CC.NumChildcareCenters,
        ft.FirefighterDepartments,
        ft.Firefighters,
        hc.ObesityRate,
        hc.AsthmaRate,
        hc.DepressionRate
    FROM target_city TC
        LEFT JOIN incometable hi ON TC.city = hi.city and TC.state
= hi.state
        LEFT JOIN policetable pt ON TC.city = pt.city and TC.state
= pt.state
        LEFT JOIN listingtable lt on TC.city = lt.city and TC.state
= lt.state
        LEFT JOIN hospitaltable h on TC.city = h.city and TC.state
= h.state
        LEFT JOIN childtable CC on TC.city = CC.city and TC.state =
CC.state
        LEFT JOIN firetable ft on TC.city = ft.city and TC.state =
ft.state
        LEFT JOIN (
                    SELECT Z.city,
                           Z.state,
                           MAX(CASE WHEN hm.Measure = 'Current
asthma among adults' THEN hm.Ratio END) AS AsthmaRate,
                           MAX(CASE WHEN hm.Measure = 'Obesity
among adults' THEN hm.Ratio END) AS ObesityRate,
                           MAX(CASE WHEN hm.Measure = 'Depression
among adults' THEN hm.Ratio END) AS DepressionRate
                    FROM zipcode Z JOIN healthmeasure hm on
Z.zipcode = hm.zipcode
                    WHERE Z.city ilike :city and Z.state ilike
:state
                    group by z.city, Z.state
```

```
                  ) AS hc On TC.city = hc.city and TC.state =
hc.state
    WHERE TC.city ilike :city and TC.state ilike :state
),

similar_city AS (
    SELECT
        sp.city,
        sp.state,
        sp.TotalPopulation,
        hi.MeanIncome,
        lt.avgmedianlistingprice,
        pt.PoliceStations,
        pt.PoliceOfficers,
        h.NumHospitals,
        CC.NumChildcareCenters,
        ft.FirefighterDepartments,
        ft.Firefighters,
        hc.ObesityRate,
        hc.AsthmaRate,
        hc.DepressionRate
    FROM similar_pop sp
        LEFT JOIN incometable hi ON sp.city = hi.city and sp.state
= hi.state
        LEFT JOIN policetable pt ON sp.city = pt.city and sp.state
= pt.state
        LEFT JOIN listingtable lt on sp.city = lt.city and sp.state
= lt.state
        LEFT JOIN hospitaltable h on sp.city = h.city and sp.state
= h.state
        LEFT JOIN childtable CC on sp.city = CC.city and sp.state =
CC.state
        LEFT JOIN firetable ft on sp.city = ft.city and sp.state =
ft.state
        LEFT JOIN (
                    SELECT Z.city,
                           Z.state,
                            MAX(CASE WHEN hm.Measure = 'Current
asthma among adults' THEN hm.Ratio END) AS AsthmaRate,
                            MAX(CASE WHEN hm.Measure = 'Obesity
among adults' THEN hm.Ratio END) AS ObesityRate,
                            MAX(CASE WHEN hm.Measure = 'Depression
among adults' THEN hm.Ratio END) AS DepressionRate
                    FROM zipcode Z JOIN healthmeasure hm on
Z.zipcode = hm.zipcode
                    group by z.city, Z.state
                    ) AS hc On sp.city = hc.city and sp.state =
hc.state
)
```

```
SELECT
    az.city,
    az.state,
    ROUND(
            ABS(COALESCE(az.TotalPopulation, 0) -
COALESCE(cz.TotalPopulation, 0)) * 0.05 +
            ABS(COALESCE(az.MeanIncome, 0) -
COALESCE(cz.MeanIncome, 0)) * 0.1 +
            ABS(COALESCE(az.avgmedianlistingprice, 0) -
COALESCE(cz.avgmedianlistingprice, 0)) * 0.1 +
            ABS(COALESCE(az.PoliceStations, 0) -
COALESCE(cz.PoliceStations, 0)) * 0.05 +
            ABS(COALESCE(az.PoliceOfficers, 0) -
COALESCE(cz.PoliceOfficers, 0)) * 0.05 +
            ABS(COALESCE(az.NumHospitals, 0) -
COALESCE(cz.NumHospitals, 0)) * 0.1 +
            ABS(COALESCE(az.NumChildcareCenters, 0) -
COALESCE(cz.NumChildcareCenters, 0)) * 0.1 +
            ABS(COALESCE(az.FirefighterDepartments, 0) -
COALESCE(cz.FirefighterDepartments, 0)) * 0.05 +
            ABS(COALESCE(az.Firefighters, 0) -
COALESCE(cz.Firefighters, 0)) * 0.05 +
            ABS(COALESCE(az.AsthmaRate, 0) -
COALESCE(cz.AsthmaRate, 0)) * 0.05 +
            ABS(COALESCE(az.ObesityRate, 0) -
COALESCE(cz.ObesityRate, 0)) * 0.05 +
            ABS(COALESCE(az.DepressionRate, 0) -
COALESCE(cz.DepressionRate, 0)) * 0.05
    ) AS similarity_score
FROM similar_city az
         JOIN current_city cz ON TRUE
ORDER BY similarity_score ASC
offset 1 rows
fetch first 2 rows only;
```

13. Simple query

This query finds the zipcode with the lowest median home in the city and state. This will be helpful for user to see which zipcode within the city has the most affordable housing.

```
WITH city_zipcodes AS (
    -- Get all zipcodes that belong to the specified city
    SELECT DISTINCT Z.zipcode, Z.city, Z.state
    FROM zipcode Z
    WHERE LOWER(Z.city) = LOWER(:city) and LOWER(Z.state) =
LOWER(:state)
),
home_prices AS (
    -- Get median home prices for zipcodes in the city
    SELECT
```

```
              r.zipcode,
              r.medianlistingprice as median_home_price,
              cz.city,
              cz.state
    FROM localmarket r
                   INNER JOIN city_zipcodes cz ON r.zipcode =
cz.zipcode
         WHERE r.medianlistingprice IS NOT NULL
           AND r.medianlistingprice > 0
     ),
     min_price_info AS (
         -- Find the minimum price and corresponding zipcode
         SELECT
             zipcode,
             median_home_price,
             city,
             state,
             ROW_NUMBER() OVER (ORDER BY median_home_price ASC) as
rank
         FROM home_prices
     )
SELECT
    zipcode,
    median_home_price,
    city,
    state,
    rank
FROM min_price_info
WHERE rank = 1;
```

2. Credentials

Using a database IDE like DataGrip, you can connect to our database with the below credentials.

Name: database-1
ENDPOINT: database-1.ctx0s2sly9f4.us-east-1.rds.amazonaws.com
Username: postgres
Password: group7roy